Integration and Kinematics

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Integration & Kinematics Practice Exercises**

A cuboid has a length of USD 8$cm, a width of USD 5$ cm, and a height of USD 3 cm. Calculate the volume of the cuboid.

The volume of a cuboid is calculated using the formula: Volume = length × width × height. Given: length = USD 8$cm, width = USD 5$ cm, height = USD 3$cm. Volume = 8 \text{ cm} \times 5 \text{ cm} \times 3 \text{ cm} Volume = 40 \text{ cm}^2 \times 3 \text{ cm} Volume = 120 \text{ cm}^3$. Therefore, the volume of the cuboid is USD 120 cm^3.

2 **[Foundational]** • **Target: Integration & Kinematics Practice Exercises**

The equation $x^3 - x - 1 = 0$ has a root near $x = 1.3$. (a) Show that the iterative formula $x_{n+1} = (x_n + 1)^{1/3}$ converges by considering $g'(x)$. (b) Starting with $x_1 = 1.3$, calculate x_2 and x_3 to 4 decimal places.

(a) Let $g(x) = (x + 1)^{1/3}$. Differentiate: $g'(x) = \frac{1}{3}(x + 1)^{-2/3} = \frac{1}{3(x+1)^{2/3}}$ Near the root $x \approx 1.3$: $g'(1.3) = \frac{1}{3(2.3)^{2/3}} \approx \frac{1}{3(1.742)} \approx 0.191$ Since \(|g'(x)| $x_1 = 1.3$ $x_2 = (1.3 + 1)^{1/3} = (2.3)^{1/3} \approx 1.3200$ $x_3 = (1.3200 + 1)^{1/3} = (2.3200)^{1/3} \approx 1.3238$

3 **[Foundational]** • **Target: Integration & Kinematics Practice Exercises**

Variables x and y satisfy $y = ax^b$. A graph of $\ln y$ against $\ln x$ is a straight line passing through $(0, 1.20)$ and $(4, 6.80)$. Find the values of the constants a and b to 3 significant figures.

Take natural logarithms of $y = ax^b$: $\ln y = \ln a + b \ln x$ This is a linear equation $Y = mX + c$ with $Y = \ln y$, $X = \ln x$, gradient $m = b$, and intercept $c = \ln a$. Gradient: $b = \frac{6.80 - 1.20}{4 - 0} = \frac{5.60}{4} = 1.40$ Vertical intercept gives $\ln a = 1.20$: $a = e^{1.20} \approx 3.32$ Therefore, $a = 3.32$ and $b = 1.40$.

4 **[Foundational]** • **Target: Integration & Kinematics Practice Exercises**

In triangle XYZ, $XY = 7$ cm, $YZ = 10$ cm, and angle $XYZ = 65^\circ$. Calculate (a) the length of XZ to 3 significant figures and (b) the area of triangle XYZ.

(a) Use the Cosine Rule: $XZ^2 = 7^2 + 10^2 - 2(7)(10)\cos(65^\circ) = 49 + 100 - 140(0.4226) = 149 - 59.166 = 89.834$ $XZ = \sqrt{89.834} \approx 9.4779\ldots \approx 9.48$ cm (3 s. f.) (b) Area of triangle:

$\text{Area} = \frac{1}{2}ab\sin C = \frac{1}{2}(7)(10)\sin(65^\circ) = 35(0.9063) \approx 31.7 \text{ cm}^2$ (3 s. f.)

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5 [Foundational] • Target: Integration & Kinematics Practice Exercises

Solve the quadratic equation $2x^2 + 5x - 4 = 0$. Give your solutions correct to 2 decimal places.

Identify coefficients: $a = 2$, $b = 5$, $c = -4$. $x = \frac{-5 \pm \sqrt{5^2 - 4(2)(-4)}}{2(2)} = \frac{-5 \pm \sqrt{25 + 32}}{4} = \frac{-5 \pm \sqrt{57}}{4}$
$x_1 = \frac{-5 + 7.5498}{4} = \frac{2.5498}{4} \approx 0.64$ $x_2 = \frac{-5 - 7.5498}{4} = \frac{-12.5498}{4} \approx -3.14$ Solutions: $x = 0.64$ or $x = -3.14$ (both to 2 d.p.).

6 [Foundational] • Target: Integration & Kinematics Practice Exercises

A car travels a distance of $d = 180$ km correct to the nearest 10 km in a time of $t = 2.4$ hours correct to 1 decimal place. Calculate the upper bound for the average speed of the car in km/h.

Find the bounds for distance and time: $d_{\text{upper}} = 180 + 5 = 185$ km $t_{\text{lower}} = 2.4 - 0.05 = 2.35$ hours To maximize speed $s = \frac{d}{t}$, divide the upper distance by the lower time:
$s_{\text{upper}} = \frac{185}{2.35} \approx 78.723\ldots \approx 78.7$ km/h (3 s. f.)

7 [Foundational] • Target: Integration & Kinematics Practice Exercises

The curve $y = f(x)$ has a stationary point at $(4, -6)$. State the coordinates of the corresponding stationary point on the curve: (a) $y = f(x + 3) + 2$, (b) $y = 3f(2x)$, (c) $y = -f(-x)$.

(a) $y = f(x + 3) + 2$ translates 3 units left and 2 units up: $(4 - 3, -6 + 2) = (1, -4)$. (b) $y = 3f(2x)$ stretches horizontally by $\frac{1}{2}$ and vertically by 3: $\left(4 \times \frac{1}{2}, -6 \times 3\right) = (2, -18)$. (c) $y = -f(-x)$ reflects in both axes (point reflection through the origin): $(-4, -(-6)) = (-4, 6)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Integration & Kinematics Practice Exercises

An infinite geometric series has common ratio $r = -0.4$ and converges to an infinite sum of $S_\infty = 25$. Find the first term a_1, and compute the exact value of the partial sum S_4.

Use the infinite sum formula $S_\infty = \frac{a_1}{1 - r}$: $25 = \frac{a_1}{1 - (-0.4)} = \frac{a_1}{1.4} \Rightarrow a_1 = 25 \times 1.4 = 35.$ Compute the 4th partial sum $S_4 = \frac{a_1(1 - r^4)}{1 - r}$: $(-0.4)^4 = 0.0256.$ $S_4 = \frac{35(1 - 0.0256)}{1.4} = \frac{35(0.9744)}{1.4} = 25 \times 0.9744 = 24.36.$

9 [Intermediate] • Target: Integration & Kinematics Practice Exercises

The table shows the time, t minutes, taken by 50 students to complete a homework task. Intervals: $0 < t \leq 10$ (freq: 8), $10 < t \leq 20$ (freq: 14), $20 < t \leq 30$ (freq: 18), $30 < t \leq 40$ (freq: 10). Calculate an estimate of the mean time taken.

Find class midpoints (x) and products (fx):Midpoint 5: $8 \times 5 = 40$Midpoint 15: $14 \times 15 = 210$Midpoint 25: $18 \times 25 = 450$Midpoint 35: $10 \times 35 = 350$ $\sum fx = 40 + 210 + 450 + 350 = 1050$
Estimated Mean $= \frac{\sum fx}{\sum f} = \frac{1050}{50} = 21$ minutes

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10 [Intermediate] • Target: Integration & Kinematics Practice Exercises

Find the 15th term and the sum of the first 20 terms of the arithmetic progression $4, 9, 14, 19, \ldots$.

Step 1: Identify progression parameters: First term $a_1 = 4$, common difference $d = 9 - 4 = 5$. Step 2: Compute a_{15}: $a_{15} = a_1 + (15-1)d = 4 + 14(5) = 4 + 70 = 74$. Step 3: Compute finite sum S_{20}:

$S_{20} = \frac{20}{2}[2a_1 + (20-1)d] = 10[2(4) + 19(5)] = 10[8 + 95] = 10(103) = 1030$.

11 [Intermediate] • Target: Integration & Kinematics Practice Exercises

A bank account pays compound interest of 3% per year. How many complete years will it take for an initial deposit of USD 2000 to exceed USD 3000?

We need $2000(1.03)^n > 3000 \Rightarrow (1.03)^n > 1.5$. Taking logarithms: $n\ln(1.03) > \ln(1.5) \Rightarrow n > \frac{\ln(1.5)}{\ln(1.03)} \approx \frac{0.405465}{0.029559} \approx 13.72$. The minimum number of complete years is 14. [M1 for setting up inequality, A1 for 14 years]

12 [Intermediate] • Target: Integration & Kinematics Practice Exercises

A student computes the volume of a cylinder using $r = 3.2$ cm and $h = 7.85$ cm. The raw calculator display shows $252.518...$. How should the final answer be reported according to Cambridge AS Level guidelines?

Unless otherwise specified, non-exact numerical answers must be given to 3 significant figures. Rounding $252.518...$ to 3 s.f. gives $253\ \text{cm}^3$. Giving 1 or 2 s.f., or excessive decimal places without justification, incurs an accuracy penalty (A1).

13 [Intermediate] • Target: Integration & Kinematics Practice Exercises

In a survey of 60 students, 35 study French, 28 study German, and 12 study both languages. Find (a) the number of students who study neither language, and (b) the probability that a student chosen at random studies French given that they study German.

(a) Inclusion-Exclusion: $n(F \cup G) = n(F) + n(G) - n(F \cap G) = 35 + 28 - 12 = 51$ $n(\text{neither}) = 60 - 51 = 9$ students (b) Conditional probability: $P(F \mid G) = \frac{n(F \cap G)}{n(G)} = \frac{12}{28} = \frac{3}{7} \approx 0.429$

14 [Intermediate] • Target: Integration & Kinematics Practice Exercises

A radioactive isotope has a decay constant $k = 0.035\ \text{year}^{-1}$. (a) Find its half-life to the nearest tenth of a year. (b) If an initial sample contains 250 mg, how much remains after 40 years?

(a) Half-life calculation: $t_{1/2} = \frac{\ln 2}{k} = \frac{0.693147}{0.035} \approx 19.8$ years. (b) Remaining mass at $t = 40$:

$A(t) = A_0 e^{-kt} = 250e^{-0.035(40)} = 250e^{-1.4} \approx 250(0.246597) \approx 61.65$ mg.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

In a manufacturing batch, 5% of components are defective. A sample of 12 components is inspected. Find the probability that at least 2 components are defective.

Let $X \sim B(12, 0.05)$. We need $P(X \geq 2) = 1 - [P(X = 0) + P(X = 1)]$. $P(X = 0) = (0.95)^{12} \approx 0.5404$. $P(X = 1) = 12(0.05)(0.95)^{11} \approx 0.3413$. Thus, $P(X \geq 2) = 1 - (0.5404 + 0.3413) = 1 - 0.8817 = 0.118$ (3 s.f.).

16 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

In a cyclic quadrilateral $ABCD$, angle $ABC = (2x + 15)^\circ$ and angle $ADC = (3x - 10)^\circ$. Find the value of x and hence the size of angle ABC. Give a geometric reason for your equation.

Geometric reason: Opposite angles of a cyclic quadrilateral sum to 180°. $(2x + 15) + (3x - 10) = 180$ $5x + 5 = 180 \Rightarrow 5x = 175 \Rightarrow x = 35$ Calculate angle ABC: $\angle ABC = 2(35) + 15 = 70 + 15 = 85^\circ$

17 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

$f(x) = cx^2$ for $0 \leq x \leq 3$, zero otherwise. Find c, $E(X)$, and $P(1 \leq X \leq 2)$.

$\int_0^3 cx^2\,dx = 9c = 1 \Rightarrow c = \frac{1}{9}$. $E(X) = \frac{1}{9}\int_0^3 x^3\,dx = \frac{1}{9} \cdot \frac{81}{4} = \frac{9}{4}$. $P(1 \leq X \leq 2) = \frac{1}{9}\left[\frac{x^3}{3}\right]_1^2 = \frac{1}{27}(8 - 1) = \frac{7}{27}$.

18 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A manufacturer claims that the mean breaking strength of cables is at least 80 kN. A sample of 16 cables gives $\bar{x} = 77.5$ kN with population SD 5 kN. Test this claim at 1% significance.

$H_0 : \mu = 80$, $H_1 : \mu < 80$. $Z = \frac{77.5 - 80}{5/4} = -\frac{2.5}{1.25} = -2$. Critical value at 1%: -2.326. Since $-2 > -2.326$, do not reject H_0. Insufficient evidence at 1% that mean breaking strength is below 80 kN.

19 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

A discrete random variable Y has $E(Y) = 5$ and $\text{Var}(Y) = 4$. Calculate $E(Y^2)$ and $\text{Var}(10 - 3Y)$.

Using $\text{Var}(Y) = E(Y^2) - [E(Y)]^2$, we get $E(Y^2) = \text{Var}(Y) + [E(Y)]^2 = 4 + 5^2 = 4 + 25 = 29$. For the linear transformation: $\text{Var}(10 - 3Y) = (-3)^2\text{Var}(Y) = 9 \times 4 = 36$.

20 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

A continuous dataset is grouped into class intervals. The interval $20 \leq x < 35$ has a frequency of 45. Find the frequency density for this class.

Class width $w = 35 - 20 = 15$. Frequency density $= \frac{\text{frequency}}{\text{class width}} = \frac{45}{15} = 3.0$. In a histogram, the height of the bar drawn on the interval $[20, 35)$ is 3.0.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Integration & Kinematics Practice Exercises

A particle moves in a straight line such that its displacement s metres from a fixed origin after t seconds is given by the equation $s = t^3 - 6t^2 + 9t + 2$. The velocity of the particle is given by $v = \frac{ds}{dt}$. Find the value of t for which the particle is instantaneously at rest and then determine the displacement of the particle at that instant.

a. t=1 s, s=6 m

b. t=2 s, s=4 m

c. **t=3 s, s=2 m ✓**

d. t=1 s, s=2 m

The displacement is given by $s = t^3 - 6t^2 + 9t + 2$. To find the velocity, we differentiate the displacement with respect to time: $v = \frac{ds}{dt} = \frac{d}{dt}(t^3 - 6t^2 + 9t + 2) = 3t^2 - 12t + 9$. The particle is instantaneously at rest when its velocity is zero, i.e., $v = 0$. So, we set the velocity equation to zero: $3t^2 - 12t + 9 = 0$. Divide by 3: $t^2 - 4t + 3 = 0$. Factorize the quadratic equation: $(t - 1)(t - 3) = 0$. This gives two possible values for t: $t = 1$ or $t = 3$. The question asks for 'the.

22 [Foundational] • Target: Integration & Kinematics Practice Exercises

The curve C has equation $y = \frac{\ln x}{x^2}, \quad x > 0$. The point M is the stationary point of C, and the line L is the tangent to C at M. The finite region R is bounded by the curve C, the tangent line L, and the vertical line $x = 1$. Find the exact area of R.

a. $\frac{4\sqrt{e} - 2e - 1}{2e}$ ✓

b. $\frac{4\sqrt{e} - 2e + 1}{2e}$

c. $\frac{2e - 3\sqrt{e}}{2e}$

d. $\frac{2\sqrt{e} - 2e - 1}{2e}$

1. **Find the stationary point M:** Using the quotient rule to differentiate $y = \frac{\ln x}{x^2}$: $\frac{dy}{dx} = \frac{\frac{1}{x} \cdot x^2 - (\ln x)(2x)}{x^4} = \frac{x(1 - 2\ln x)}{x^4} = \frac{1 - 2\ln x}{x^3}$. Setting $\frac{dy}{dx} = 0$: $1 - 2\ln x = 0 \Rightarrow \ln x = \frac{1}{2} \Rightarrow x = e^{1/2} = \sqrt{e}$. The y-coordinate at M is: $y = \frac{\ln(\sqrt{e})}{(\sqrt{e})^2} = \frac{\frac{1}{2}}{e} = \frac{1}{2e}$. Since M is a stationary point, the tangent line L is horizontal wit.

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23 [Foundational] • Target: Integration & Kinematics Practice Exercises

Two particles, A and B, move in the xy-plane. Their position vectors at time t seconds, for $t \geq 0$, are given by $\mathbf{r}_A(t) = (2t; 4t - 1)$ and $\mathbf{r}_B(t) = (3 + 5t; 1 - 2t)$ respectively. The particles start their motion at $t = 0$. Find the minimum distance between particle A and particle B.

a. $\sqrt{26}$ ✓

b. $2\sqrt{13}$

c. $3\sqrt{10}$

d. $4\sqrt{5}$

The position vectors of particle A and particle B at time t are given by $\mathbf{r}_A(t) = (2t; 4t - 1)$ and $\mathbf{r}_B(t) = (3 + 5t; 1 - 2t)$. The vector representing the displacement from particle B to particle A at time t is $\mathbf{d}(t) = \mathbf{r}_A(t) - \mathbf{r}_B(t) = (2t - (3 + 5t); (4t - 1) - (1 - 2t)) = (-3t - 3; 6t - 2)$. The distance between the particles at time t is the magnitude of this displacement vector.

24 [Foundational] • Target: Integration & Kinematics Practice Exercises

The region R in the first quadrant is bounded by the curve $y = \frac{6}{\sqrt{3x+4}}$, the coordinate axes, and the vertical line $x = k$, where $k > 0$. Given that the total area of R is 8, and the vertical line $x = c$ (where $0 < c < k$) divides R into two regions of equal area, find the exact value of c.

a. $\frac{5}{3}$ ✓

b. 2

c. $\frac{7}{3}$

d. $\frac{13}{27}$

Step 1: Integrate the function using the reverse chain rule. $\int 6(3x+4)^{-\frac{1}{2}}\,dx = \frac{6}{\frac{1}{2}\cdot 3}(3x+4)^{\frac{1}{2}} = 4\sqrt{3x+4}$
Step 2: Find the parameter k using the given total area of 8.
$\text{Area} = \int_0^k \frac{6}{\sqrt{3x+4}}\,dx = \left[4\sqrt{3x+4}\right]_0^k = 4\sqrt{3k+4} - 4\sqrt{4} = 4\sqrt{3k+4} - 8$ Setting this equal to 8:
$4\sqrt{3k+4} - 8 = 8 \Rightarrow 4\sqrt{3k+4} = 16 \Rightarrow \sqrt{3k+4} = 4$ Squaring both.

25 [Foundational] • Target: Integration & Kinematics Practice Exercises

A particle's velocity v m/s at time t seconds is given by $v = \frac{1}{2}t^2 - 3t + 5$. The displacement of the particle from a fixed origin is s metres. It is known that at $t = 2$, the displacement $s = 7$. The displacement function $s(t)$ can be expressed in the form $s(t) = At^3 + Bt^2 + Ct + D$. Find the value of A and the value of D.

a. A = 1/6, D = 5/3

b. **A = 1/6, D = 10/3** ✓

c. A = 1/3, D = 5/3

d. A = 1/3, D = 10/3

We are given the velocity function $v(t) = \frac{1}{2}t^2 - 3t + 5$. Since displacement s is the integral of velocity with respect to time, we have: $s(t) = \int v(t)\,dt = \int \left(\frac{1}{2}t^2 - 3t + 5\right)dt$ Integrating term by term, we get:
$s(t) = \frac{1}{2}\cdot\frac{t^3}{3} - 3\cdot\frac{t^2}{2} + 5t + K\, s(t) = \frac{1}{6}t^3 - \frac{3}{2}t^2 + 5t + K$ Comparing this with the given form
$s(t) = At^3 + Bt^2 + Ct + D$, we can identify $A = \frac{1}{6}, B = -\frac{3}{2}, C = 5$. The con.

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26 [Foundational] • Target: Integration & Kinematics Practice Exercises

A particle moves along a straight line. Its velocity, v m/s, at time t seconds is given by $v = 3t - 2$. Find the displacement of the particle during the first 4 seconds of its motion.

a. 16 m

b. **20 m** ✓

c. 24 m

d. 28 m

The displacement of the particle can be found by integrating the velocity function with respect to time. The displacement s is given by: $s = \int v\,dt$ In this case, $v = 3t - 2$. $s = \int (3t - 2)\,dt$ Integrating, we get: $s = \frac{3}{2}t^2 - 2t + C$ To find the displacement during the first 4 seconds, we evaluate the definite integral from $t = 0$ to $t = 4$. We can assume $C = 0$ for calculating the change in displacement from $t = 0$. $s(4) - s(0) = \left[\frac{3}{2}t^2 - 2t\right]_0^4$.

27 [Foundational] • Target: Integration & Kinematics Practice Exercises

Find the exact value of the definite integral $\int_0^{\ln 2} e^{2x}\,dx$.

a. $\frac{3}{2}$ ✓

b. 3

c. $\frac{1}{2}$

d. $\frac{7}{2}$

To evaluate the definite integral: 1. Find the antiderivative using the standard standard result $\int e^{kx}\,dx = \frac{1}{k}e^{kx} + C$: $\int e^{2x}\,dx = \frac{1}{2}e^{2x}$ 2. Apply the limits from $x = 0$ to $x = \ln 2$: $\left[\frac{1}{2}e^{2x}\right]_0^{\ln 2} = \frac{1}{2}e^{2\ln 2} - \frac{1}{2}e^{2(0)}$ 3. Simplify the exponential terms using the power law of logarithms, $2\ln 2 = \ln(2^2) = \ln 4$.

28 [Foundational] • Target: Integration & Kinematics Practice Exercises

Consider two vectors, $\mathbf{u} = (3; k)$ and $\mathbf{v} = (-2; 4)$. If $\mathbf{u}$ is parallel to $\mathbf{v}$, find the value of the scalar parameter k.

a. **-6** ✓

b. 6

c. 3

d. -3

Two vectors are parallel if one is a scalar multiple of the other. Therefore, there must exist a scalar c such that $\mathbf{u} = c\mathbf{v}$. This gives us the equations: $3 = c(-2)$ $k = c(4)$ From the first equation, we can solve for c: $c = \frac{3}{-2} = -\frac{3}{2}$ Now substitute this value of c into the second equation to find k: $k = \left(-\frac{3}{2}\right)(4)$ $k = -\frac{12}{2}$ $k = -6$ Therefore, the value of k is -6. A common algebraic trap is to mistakenly set the components equal with.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Integration & Kinematics Practice Exercises**

Consider two points in the Cartesian plane, $P = (2, 3)$ and $Q = (5, 7)$. What is the component form of the vector $\vec{PQ}$?

a. $\langle 3, 4 \rangle$ ✓

b. $\langle -3, -4 \rangle$

c. $\langle 7, 10 \rangle$

d. $\langle 1, 4 \rangle$

To find the component form of a vector $\vec{PQ}$ from point $P(x_1, y_1)$ to point $Q(x_2, y_2)$, we subtract the coordinates of the initial point P from the coordinates of the terminal point Q. The formula is $\vec{PQ} = \langle x_2 - x_1, y_2 - y_1 \rangle$. Given $P = (2, 3)$ and $Q = (5, 7)$, we have $x_1 = 2$, $y_1 = 3$, $x_2 = 5$, and $y_2 = 7$. Applying the formula: $\vec{PQ} = \langle 5 - 2, 7 - 3 \rangle$ $\vec{PQ} = \langle 3, 4 \rangle$ Therefore, the component form of the vector $\vec{PQ}$ is.

30 **[Intermediate]** • **Target: Integration & Kinematics Practice Exercises**

Two vectors are given by $\vec{u} = 3\hat{i} + k\hat{j}$ and $\vec{v} = -\hat{i} + 4\hat{j}$. If $\vec{u}$ and $\vec{v}$ are orthogonal, find the value of the scalar parameter k.

a. 3/4

b. **4/3** ✓

c. -3/4

d. -4/3

Two vectors $\vec{u}$ and $\vec{v}$ are orthogonal if and only if their dot product is zero. The dot product of $\vec{u} = u_1\hat{i} + u_2\hat{j}$ and $\vec{v} = v_1\hat{i} + v_2\hat{j}$ is given by $\vec{u} \cdot \vec{v} = u_1v_1 + u_2v_2$. Given $\vec{u} = 3\hat{i} + k\hat{j}$ and $\vec{v} = -\hat{i} + 4\hat{j}$, their dot product is: $\vec{u} \cdot \vec{v} = (3)(-1) + (k)(4)$ $\vec{u} \cdot \vec{v} = -3 + 4k$ For the vectors to be orthogonal, their dot product must be zero: $-3 + 4k = 0$ $4k = 3$.

31 **[Intermediate]** • **Target: Integration & Kinematics Practice Exercises**

Consider the vector $\mathbf{v} = \langle 3, 4 \rangle$. What is the magnitude of this vector?

a. **5** ✓

b. 7

c. 12

d. 25

The magnitude of a vector $\mathbf{v} = \langle x, y \rangle$ is given by the formula $|\mathbf{v}| = \sqrt{x^2 + y^2}$. For the given vector $\mathbf{v} = \langle 3, 4 \rangle$, we have x = 3 and y = 4. Substituting these values into the formula: $|\mathbf{v}| = \sqrt{3^2 + 4^2}$ $|\mathbf{v}| = \sqrt{9 + 16}$ $|\mathbf{v}| = \sqrt{25}$ $|\mathbf{v}| = 5$ Therefore, the magnitude of the vector $\mathbf{v}$ is 5.

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32 [Intermediate] • Target: Integration & Kinematics Practice Exercises

Consider the vector $\mathbf{v}$ with initial point $P = (2, 3)$ and terminal point $Q = (5, 7)$. What are the components of the vector $\mathbf{v}$?

a. ■3, 4■ ✓

b. ■7, 10■

c. ■-3, -4■

d. ■4, 3■

The components of a vector from point $P = (x_1, y_1)$ to point $Q = (x_2, y_2)$ are given by $\langle x_2 - x_1, y_2 - y_1 \rangle$. In this case, $P = (2, 3)$ and $Q = (5, 7)$. Therefore, the components of vector $\mathbf{v}$ are $\langle 5 - 2, 7 - 3 \rangle = \langle 3, 4 \rangle$.

33 [Intermediate] • Target: Integration & Kinematics Practice Exercises

Using the trapezium rule with 3 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 1, y_1 = 1, y_2 = 2, y_3 = 10$.

a. $\frac{17}{2}$ ✓

b. 7

c. $\frac{103}{12}$

d. $\frac{19}{2}$

$\int y\,dx \approx \frac{h}{2}[y_0 + y_3 + 2(y_1 + \cdots + y_2)] = \frac{1}{2} \times 17 = \frac{17}{2}$.

34 [Intermediate] • Target: Integration & Kinematics Practice Exercises

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = t^2 - 4t$ m/s. Find when the particle is momentarily at rest.

a. $t = 3$ s (excluding $t = 0$)

b. $t = 5$ s (excluding $t = 0$)

c. $t = 0$ s only

d. $t = 4$ s (excluding $t = 0$) ✓

Setting $v = 0$: $t^2 - 4t = 0 \Rightarrow t(t - 4) = 0 \Rightarrow t = 4$ s (excluding $t = 0$)

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35 [Intermediate] • Target: Integration & Kinematics Practice Exercises

A particle has initial velocity $u = 10$ m/s and constant acceleration $a = 4$ m/s^2. Find its velocity after $t = 3$ s.

a. **22** ✓

b. 21

c. 23

d. 14

Applying Kinematics principles, $v = u + at = 10 + (4)(3) = 22$ m/s Therefore, the verified correct option is 22.

36 [Intermediate] • Target: Integration & Kinematics Practice Exercises

Find $\int x(x^2 + 1)^2\,dx$ using substitution.

a. $\frac{x^2(x^2+1)^2}{2} + C$

b. $\frac{x^6}{6} + \frac{x^4}{2} + \frac{x^2}{2} + C$ ✓

c. $\frac{2(x^2+1)^3}{3} + C$

d. $\frac{(x^2+1)^3}{3} + C$

Let $u = x^2 + 1$, $du = 2x\,dx$: $\int x(x^2+1)^2 dx = \frac{x^6}{6} + \frac{x^4}{2} + \frac{x^2}{2} + C$.

37 [Intermediate] • Target: Integration & Kinematics Practice Exercises

A particle has initial velocity $u = 3$ m/s and constant acceleration $a = 3$ m/s^2. Find its speed after travelling a displacement $s = 11$ m.

a. $\sqrt{79}$

b. $\sqrt{71}$

c. 75

d. $\sqrt{75}$ ✓

$v^2 = u^2 + 2as = (3)^2 + 2(3)(11) = 75 \Rightarrow v = \sqrt{75}$ m/s

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A particle has initial velocity $u = 4$ m/s and constant acceleration $a = 4$ m/s^2. Find its displacement after $t = 6$ s.

a. 24

b. 98

c. 94

d. **96** ✓

$s = ut + \frac{1}{2}at^2 = 4(6) + \frac{1}{2}(4)(6)^2 = 96$ m

39 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

A particle moves so that its velocity at time t s is $v = 8t + 3$ m/s. Find the acceleration at $t = 0$ s.

a. 13

b. 10

c. **8** ✓

d. 6

$a = \frac{dv}{dt} = 8$. At $t = 0$: $a = 8$ m/s^2

40 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

A particle accelerates uniformly from $u = 2$ m/s to $v = 14$ m/s in $t = 6$ s. Find its acceleration.

a. 1

b. 3

c. **2** ✓

d. 12/7

$a = \frac{v - u}{t} = \frac{14 - (2)}{6} = 2$ m/s^2

41 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

Evaluate $\int_0^1 xe^x\,dx$ using integration by parts.

a. -1

b. **1** ✓

c. $1 + e^2$

d. 2

$\int_0^1 xe^x\,dx = 1$.

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42 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

A particle moves so that its velocity at time t s is $v = 2t$ m/s. Find an expression for its displacement s in terms of t.

a. $3t^2 + C$

b. $t^2 + C$ ✓

c. 1+C

d. $2t^2 + C$

$s = \int v\,dt = t^2 + C$

43 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

On a histogram, a class interval is $20 \leq x < 30$ with a frequency of 25. What is the frequency density for this bar?

a. **2.50** ✓

b. 5.00

c. 25

d. 10

Class width = $30 - 20 = 10$. Frequency density = $\frac{\text{Frequency}}{\text{Class Width}} = \frac{25}{10} = 2.50$. [M1 A1]

44 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

The diagram shows a number line with points marked. What is the value of x?

a. -5

b. **-3** ✓

c. 1

d. 5

The number line shows integer values from -6 to 6. Point x is located exactly at the mark for -3. Therefore, the value of x is -3.

45 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

The equation $y = 2x^2 - 8$ crosses the x-axis at:

a. $x = \pm 2$ ✓

b. $x = \pm 4$

c. $x = 2$

d. $x = \pm\sqrt{8}$

Applying Algebra & Graphs principles, $2x^2 = 8 \Rightarrow x = \pm 2$ Therefore, the verified correct option is $x = \pm 2$.

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